

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2278

Scenic Woods 100 kV
October 01, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2278. The Interconnection Customer proposes a 232.8 MW natural gas combined cycle facility interconnecting at the Scenic Woods 100 kV substation in the Indiana area. The request was received on January 16, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 534 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,248,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SCENIC WOODS / HIGH FALLS - CROGHAN NY 100.0 kV line	TWIN BRANCH PV / PARKROSE 100.0 kV L	141.6	321	100.0
PARKROSE / TWIN BRANCH PV 100.0 kV	TWIN BRANCH SUBSTATION / KEMPER COUNTY 100.0 kV	119.9	412	47.3
DENVER TERMINAL / TORTOISE 100.0 kV	TORTOISE / DENVER TERMINAL 100.0 kV Circuit 2	115.2	323	23.8
TORTOISE SUBSTATION / DENVER TERMINAL 100.0 kV	DENVER TERMINAL / TORTOISE 100.0 kV L	115.2	323	23.8
TEELAND / TORTOISE 100.0 kV Cir. 1	TEELAND / TORTOISE 100.0 kV Circuit 2	113.5	351	24.1
TORTOISE / TEELAND SUBSTATION 100.0 kV Line 2	TEELAND / TORTOISE 100.0 kV	113.5	351	24.1
TEELAND / TABLE MT 100.0 kV Cir. 1	TEELAND / TABLE MT 100.0 kV L	112.0	391	25.2
TABLE MT / TEELAND 100.0 kV Circuit 2	TEELAND / TABLE MT 100.0 kV Circuit 1	112.0	391	25.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SCENIC WOODS / HIGH FALLS - CROGHAN NY 100.0 kV line	Rebuild line with higher-capacity conductor	\$1,738,000	48% (\$825,000)
PARKROSE / TWIN BRANCH PV 100.0 kV	Rebuild line with higher-capacity conductor	\$4,587,000	23% (\$1,060,000)
DENVER TERMINAL / TORTOISE 100.0 kV	Rebuild line with higher-capacity conductor	\$2,321,000	12% (\$282,000)
TORTOISE / DENVER TERMINAL 100.0 kV	Rebuild line with higher-capacity conductor	\$2,641,000	12% (\$321,000)
TEELAND / TORTOISE 100.0 kV	Rebuild line with higher-capacity conductor	\$1,634,000	12% (\$199,000)
TEELAND SUBSTATION / TORTOISE 100.0 kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,888,000	12% (\$230,000)
TEELAND SUBSTATION / TABLE MT 100.0 kV	Rebuild line with higher-capacity conductor	\$1,416,000	12% (\$172,000)
TEELAND / TABLE MT 100.0 kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,303,000	12% (\$159,000)
	Total allocated to project		\$3,248,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.